SoundMagic Spectral

Real-time spectral plug-ins for creative audio processing

User Documentation – Michael Norris, Te Kōkī New Zealand School of Music
michael.norris@nzsm.ac.nz • http://www.michaelnorris.info/soundmagicspectral
This revision: © June 2016

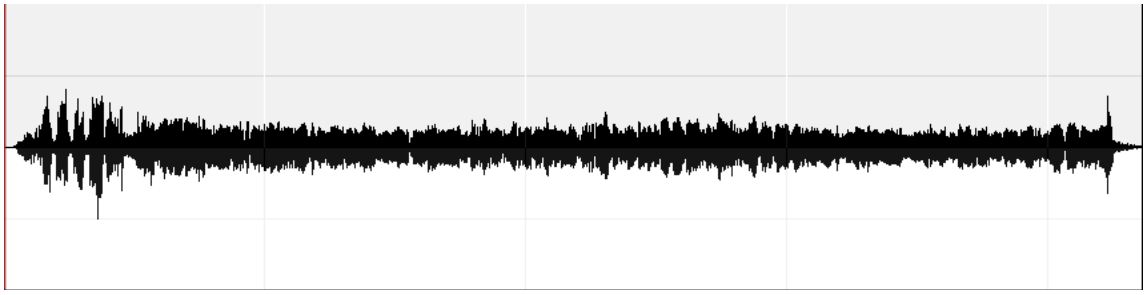
1. Introduction

SoundMagic Spectral is a suite of freeware Audio Unit plug-ins that offer exotic real-time spectral processing effects, along with a few time-domain effects. This document explains the theory behind spectral processing and details each plug-in in the suite.

2. Understanding spectral processing

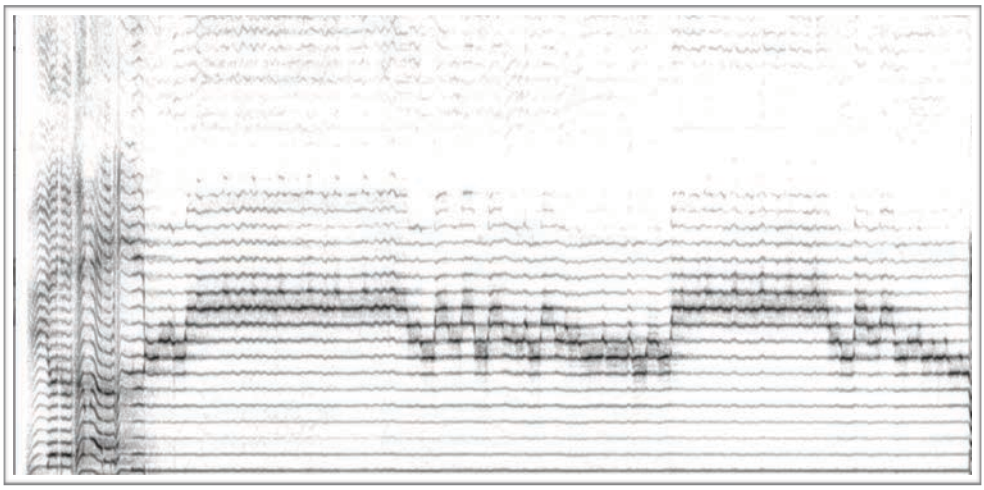
2.1. Time domain vs. frequency domain processing

A time domain representation of sound is one in which only the instantaneous changes in sound pressure (amplitude) are displayed, recorded or manipulated. This is how samples are normally stored in a soundfile, and is how waveform editors display the sound graphically:



While this graphic tells us about the overall volume changes in the sound, it doesn't tell us much about what frequencies are in the sound – for all we know, this could be high-frequency or low-frequency.

In the next image, however, we can see the same sound in a **FREQUENCY DOMAIN REPRESENTATION** (or 'spectral representation'), where the x-axis=time, the y-axis=frequency, and darkness=amplitude.



Here the raw sound data has been **converted** into a series of frequencies and amplitudes. Towards the top of the graphic are the high frequencies, and the low frequencies are near the bottom. The amplitudes of individual components are represented by the darkness of the frequencies. In the soundfile represented above, we can see that while the overtones of the spectrum remain constant in frequency (the stripes are pretty flat, except in the opening segment), different harmonics are being highlighted. In fact, this soundfile is a recording of Tuvan throatsinging, and the spectrogram graphically shows how the throatsinger is managing to create a melodic line even without changing the fundamental frequency of his vocal cords. This sort of information could not be seen at all in the time domain representation.

Apart from using the frequency domain to display more information about a sound, we can also the frequency domain to transform the sounds in ways that are impossible in the time domain. We can alter specific frequencies, we can stretch out and smooth over amplitude changes, and generally it gives us a lot of creative possibilities.

Common to all spectral effects are two fundamental algorithms: **spectral analysis** (converting time domain → frequency domain) and **spectral synthesis** (frequency domain → time domain). In order for a spectral effect to operate in real-time, this analysis and resynthesis must happen on the order of 150 to 200 times *every second* for a typical configuration.

2.1. The Fast Fourier Transform & real-time spectral processing

The algorithm that drives these conversions is called the **FAST FOURIER TRANSFORM** (FFT for short). Compared to most time-domain algorithms, FFTs are quite computationally expensive: back in the 1990s, when SoundMagic was first written, it was not possible to run FFTs in real-time, except on dedicated hardware boards that had to be installed in your computer. By the early-to-mid-2000s, however, there was a rapid increase in processor capability, and companies such as Apple gave programmers access to hardware-optimized maths routines, including the FFT . In short, suddenly real-time spectral processing became possible.

2.1. Spectral bins and FFT sizes

In order to understand how these effects work, it's useful to have a little bit of 'behind the scenes' understanding of the FFT. The FFT can only work with 'chunks' of audio at a time, and these chunks must contain a set number of samples which is a power of 2 (most commonly 1024 or 2048). If we say that the FFT size is 'n', then the FFT algorithm converts the n samples into an array of n/2 'bins', where each bin covers the 'energy' contained within a range of frequencies which are equally spaced across the spectrum from 0Hz up to the Nyquist limit (half the sampling rate). For instance, if you chose an FFT size of 1024 and assuming a sample rate of 44100Hz, then the FFT will convert 1024 samples of audio into 512 'bins', each bin covering 22050/512 = c. 43Hz. (So Bin 0 goes from 0–43Hz, Bin 1 goes from 43–86Hz, etc)

As you can see, Bin 0 includes quite a lot of low frequencies, which means that it doesn't have great 'frequency resolution'. It might seem like the obvious solution is to set the FFT size higher – but there is a tradeoff between the amount of frequency data that can be captured (the higher, the better) and the time accuracy upon resynthesis (the smaller, the better). When FFTs are resynthesised, if their phases and amplitudes have been changed (which they usually have been in spectral effects), the timings of these frequencies can subtly drift, which creates an aural effect known as 'phasiness', a characteristic sound of many FFT effects. So you have to balance the FFT size to the particular application – there's really no rule-of-thumb other than trial-and-error. As a general observation, however, effects that deliberately 'smooth' or 'smear' frequencies (such as Spectral Dronemaker) sound smoother and more luscious with large FFT sizes such as 16384 (or even larger if your CPU can handle it).

3. A GUIDE TO THE SOUNDMAGIC SPECTRAL PROJECT

The SoundMagic Spectral project builds on the ideas implemented in the SoundMagic plug-in suite that I began in 1995 for the now discontinued OS 9 application SoundEffects (www.riccisoft.com/soundeffects), later called SoundMaker. SoundMagic Spectral brings not only effects familiar from the SoundMagic suite to Mac OS X through the standard plug-in format called 'Audio Units', but furthermore adds many new features unthinkable in SoundMaker: real-time processing, hardware-optimized FFT and phase vocoder routines and even more exotic and creative effects than in the original suite.

The concepts behind the SoundMagic algorithms owe much to the work of previous composers and programmers – most notably Trevor Wishart in his book Audible Design (a must-read for those using DSP) based upon his work with the Composers Desktop Project.

3.1. Downloading

These effects may be downloaded from the following URL: http://www.michaelnorris.info/soundmagicspectral. Note that they have an 'eternal beta' status, meaning that I offer very little in the way of technical support – I can only really deal with installation issues, and that's about it. These effects are sporadically updated, but as I am a father of three and have a full-time job as an academic, I find it difficult to do much more than occasional bug fixes and format updates.

3.2. Licence and support

The current set of plug-ins are offered as freeware. It's entirely possible that the effects contain some lingering bugs/quirks, though hopefully I've ironed most of them out. While I am open to, and interested in, suggestions for new effects and new parameters for existing effects, I can't guarantee immediate satisfaction of your request. I do have a working to-do list, however, so email through any bugs/suggestions to me at <*michael.norris@nzsm.ac.nz*>.

3.3. Minimum Technical Requirements

SoundMagic Spectral requires Mac OS X 10.6 or later. An AudioUnit-compatible host, such as Logic, Performer, Live, Amadeus, Sound Studio, or Max/MSP using my au~ object, is needed.

3.4. Latency & tails

Because these effects use large chunks of sound, there is often significant latency, usually of about half a second, sometimes longer, between the input and output. This means that it's a good idea to ensure there is at least a second's worth of silence at the end of your sound file, to allow for the processing 'tail'.

3.5. Phase Randomization

The optimized phase vocoder routines in these effects do not perform the usual 'phase unwrapping' stage that is typically seen in most phase vocoder implementations. The reason is that it is computationally expensive, and for most processes is actually redundant. By removing it, I have made about a 10–15% saving in CPU utilization of these effects. The downside is that when you apply certain procedures such as amplitude interpolation, you tend to get a comb filtering/ringing effect, at a frequency at the size of the FFT. To combat this, I have implemented a simple trick, which is to randomize the phase values of each frequency component. While this removes the ringing effect, as the phases no longer coincide at each FFT period, it also removes the temporal accuracy of phase components, which tends to 'smear' the sound, especially at large FFT sizes. On the plus side, this can

be quite sonically desirable for many applications, and so you will often see this option selected by default.

3.1. SPECTRAL CLICHÉS

As much as I love playing with these effects, it's only right that I warn you about 'spectral soup' – a very easy-to-create spectral cliché of arbitrary sine tones running up and down the spectrum. The expressive power of these effects is in their more subtle manipulations of timbre. Often the most rewarding effects are found when you move a parameter just slightly away from its 'normal' position. And don't forget you chain these effects in series for even more possibilities!

---

4. THE EFFECTS

I have categorized the effects in SoundMagic Spectral below by their typical function or application. I have divided them into the following categories:

- Spectral Smoothing Effects
- Spectral Sustaining Effects
- Spectral Excitement Effects
- Spectral Pitch Effects
- Spectral Filtering Effects
- Spectral Texturizing Effects
- Spectral Partial Effects
- Time Domain Effects

Common Parameters

- Brightness
  Applies a filter to either increase or decrease the strength of higher-frequency components. At its most aggressive, the upper frequency components can 'sizzle' quite audibly.
- Lo bin cutoff
  Frequency components lower than a certain frequency are removed (set to zero amplitude)
- Hi bin cutoff
  Frequency components higher than a certain frequency are removed (set to zero)
- Randomize phases
  The phase information is randomized (see 'Phase Randomization' section above)
- FFT size
  The larger the FFT size, the 'smoother' the result, but also the more CPU is used in performing the FFT. On G5s, this probably isn't an issue, but on G4s, going too high can cause the audio to break up as the CPU maxes out.
- Gain
  The output level of the effect. Some 'accumulation' effects can create clipping in the audio – reducing the gain will avoid this.
- Feedback
  The amount with which the output of the effect is 'fed back' into the input
- Parameter Variance
  Some effects allow you to alter a parameter in a programmatic way, through the notion of 'variance'. This lets you 'connect' one parameter to either the amplitude of the sound, the inverse of the amplitude, a Low Frequency Oscillator (LFO) or to a random change.

5. SPECTRAL SMOOTHING EFFECTS

The effects in this category tend to *blur*, *smooth* or otherwise slow down the rate of spectral change. The net result is that one can create strange hybrid sounds, or *drones* that are still based on the sonic characteristics of the original.

SPECTRAL AVERAGING

Averages the spectral information over n windows. Very similar effect to Spectral Blurring.

- Number of Frames
  The number of FFT frames over which to average. The more frames the smoother the result.

SPECTRAL BLURRING

Applies a low-pass filter to the spectral changes from one window to the next, progressively 'smearing' the sound. Has similar spurious problems as Spectral Averaging, but may still be useful.

- Blur amount
  The amount of 'smearing' that takes place. This can be varied using the variance controls detailed above

SPECTRAL DRONEMAKER

Spectral DroneMaker is the 'king' of spectral drone effects. It can turn any sound into a beautiful, slowly-changing drone based on the sonic characteristics of the underlying sound. The idea came from the software called *Mammut* developed by Øyvind Hammer of Notam, which used very large FFTs to 'smear' out the detail, thus creating slowly changing drones. Unlike Mammut, Spectral DroneMaker still uses normal-sized FFTs, instead creating the drone by gradually interpolating between frequency components sampled at a delay. Other parameters for making the drone even more sexy include a bank of comb filters added prior to the interpolation for a subtle, or not-so-subtle pitched effect, the ability to gate frequency components as in Spectral Tracing, and the ability to add a flanger to the output.

- Interpolation length
  Each bin is interpolated between samples taken at a distance. This parameter sets the amount of time that the bins are sampled at.
- Interp length variance
  The above parameter can be randomized within a range of interp length ± variance
- Use peak amplitudes
  With this switched on, the process will scan for the highest amplitude that occurs in each bin the input file while they are being interpolated. The process will then use that stored amplitude instead of the actual value that occurs at the next interpolation point in the input file.
- Gate level
  Only lets partials above a certain threshold into the output sound.
- Comb filterbank level
  Spectral DroneMaker allows you to "harmonize" the input sound by putting it through a comb filterbank. This parameter sets the wet/dry level of the comb filtering.
- Comb filter fundamental
  Sets the fundamental frequency of the filterbank. The filters are built up from this frequency.
- Scale type
  A list of preset scale types for building the filterbank
- Number of octaves
  How many octaves you want the filterbank to cover. The filterbank can be quite computationally expensive with a high number of octaves.
- Filter resonance
  The strength of the comb filters. Watch out for distortion from high resonance filters.
- Flanger amount
  Lets you apply a time-varying flanger to the output
- Flanger depth
  The maximum delay length of the flanger. The higher the number the more "intense" the pitch-varying aspect of the flanger
- Flanger rate
  The speed of the flanger.

6. SPECTRAL SUSTAINING EFFECTS

These effects in some way sustain partials for a certain duration, using slightly different algorithms to determine when and for how long partials should be sustained for.

SPECTRAL FREEZING

Sometimes also known as 'Spectral Accumulation'. Each bin is watched until its amplitude reaches a peak. It is then held at that peak until it is exceeded by another peak in that bin. The Freeze Factor parameter puts a decay onto the peak holding, so that you don't just end up with a whole series of very high-amplitude peaks.

- Freeze Factor
  Once a bin is frozen, it can then be 'decayed'. If this is set to 100%, no decaying takes place, allowing for a complete bin freeze, which will only be superceded by a peak.
- Use threshold param
  If this is checked, a freeze will only take place if the new peak is above a threshold level.
- Threshold
  The threshold above which a freeze can take place
- Zero bins under threshold
  If this is checked, unfrozen bins under the threshold level will be zeroed

SPECTRAL GATE AND HOLD

A similar effect to Spectral Freezing, in which each bin is watched until its amplitude exceeds a threshold level. The main difference between this effect and Spectral Freezing, is that a bin is frozen for a specific length of time, rather than until it is exceeded by a higher amplitude. In Spectral Gate and Hold, once a bin is frozen, it is held for a specific duration (indicated by the Hold duration and Hold variance parameters). During that time there is the possibility of an optional decay (indicated by the Decay factor parameter).

- Threshold
  The level above which a bin will be frozen
- Hold duration
  The length of time to hold a frozen bin for
- Hold variance
  The "hold duration" parameter can be randomized on a per-bin basis within a range of hold duration ± var
- Decay factor
  The amount that the frozen bin is decayed by over its lifetime. 100% = no decay, 0% = instantaneous decay
- Zero partials under threshold
  Partial falling under the threshold will be zeroed. If this is checked, then nothing of the original sound will come through, except the frozen bins.

SPECTRAL WEAVE

This effect waits until the audio crosses a threshold and is sustained above that threshold, and then samples one FFT frame. The effect keeps this frame in memory for a set length of time, while creating sustained 'threads' of sound from the frame by sustaining in and applying a fade in-fade out envelope.

A number of independent threads can be held in memory simultaneously, so that you get a 'contrapuntal weaving' of spectral sustains.

- Threshold
  The dynamic level above which the audio must rise before a thread is captured.
- Num threads
  The maximum number of threads to hold in memory at one time.
- Thread life
  How long a thread exists in memory until it is cleared and space is made for a new thread.
- Max stability
  How stable the audio must be above the threshold for a thread to be captured.
- Min stability length
  The minimum length of time for which the audio must be held within the stability limits above before a thread is captured.
- Hold length & Hold length variance
  The total duration of each 'fade in - fade out' envelope ± the variance.
- Fade in/out %
  The % of the hold length to which the fade curve is applied. 100% means that the first half of the sound is a fade in, and the second half is a fade out.
- Silence length & Silence length var
  The duration of silence (± the variance) between each fade in/out envelope.
- Hold out length
  After a thread has been captured, how long the effect should 'hold out' until it starts listening to the audio again to capture another thread.
- Pulse frequency & pulse gain
  A simple amplitude modulation applied to the fade in/out envelope.

7. SPECTRAL EXCITEMENT EFFECTS

SPECTRAL SHIMMER

A variety of ways to add "shimmer", "jitter", "sparkle", "pulsation" or "sizzle" to your sound.

- Jitter amount
  The percentage of bins that are "jittered" (a random fluctuation in amplitude)
- Jitter only above
  Allows jitter only above a certain frequency
- Low-frequency jitter
  The maximum amount of jitter (in dB) at 0Hz
- High-frequency jitter
  The maximum amount of jitter (in dB) at the Nyquist limit. The amount of jitter is interpolated between this value and the low-frequency jitter over the range.
- Jitter gate level
  Only apply jitter if the bin is above a certain amplitude (a low level here can create a real "sizzle" in the higher frequencies)
- Jitter hold (windows)
  The number of windows for which a jitter is "held" on a bin – a larger value makes less of a "jitter" and more of a kind of "surging" sound

- Amplitude pulse type
  Applies an amplitude pulsation wave across the spectrum. "Square" uses blocks of bins (of size set by the "Pulse Width" parameter), either of max or min pulse level. "Sin" uses a sin wave across the spectrum ranging between max & min levels.
- Pulse level max
  The maximum amplification of the pulse wave
- Pulse level min
  The minimum amplification of the pulse wave
- Pulse width
  The number of bins that a pulsation wave is spread across
- Pulse rate
  The rate at which the pulsation wave is cycled up or down across the spectrum
- Delay variance
  Delays each bin in a programmatic way, creating sweeps up or down or random delay fluctuations. The Delay Variance parameter sets the maximum delay, while the Delay Type parameter sets the particular method by which bins are delayed.

SPECTRAL SHUFFLE

Randomly shuffles blocks of bins around within the spectrum. You have control over the Shuffle factor (the amount of blocks that are shuffled), the Shuffle range (how far away from the original position each block is moved) and the number of bins per block.

- Shuffle factor
  The number of groups that are shuffled, as a percentage of the FFT size.
- Shuffle range
  How far away, in bins, each group is moved from its original position.
- Bins per group
  The number of bins that are kept together in a block.

SPECTRAL PULSING

A rather experimental effect, that switches bins on and off across the spectrum in a pulsating fashion.

- On duration
  The length of time to keep bin zero switched on
- Off duration
  The length of time to keep bin zero switched off (after the on duration)
- On multiplier freq
  As you go up the frequency spectrum, multiply the on duration by this number
- Off multiplier freq
  As you go up the frequency spectrum, multiply the off duration by this number
- On multiplier time
  As you move forward in time, multiply the on duration by this number
- Off multiplier time
  As you move forward in time, multiply the off duration by this number
- Min on duration
  Set the minimum length for the on duration
- Max on duration
  Set the maximum length for the on duration
- Min off duration

Set the minimum length for the off duration

- Max off duration
  Set the maximum length for the off duration
- Modulo type
  1= ping-pong between min and max; 2 = wrap round from max to min/min to max

8. SPECTRAL PITCH EFFECTS

SPECTRAL HARMONIZER

The spectrum is transposed up or down by a set series of intervals (you have three to play with) and mixed back into the original. Each interval can then be transposed in a similar manner to the Spectral Filterbank transpositions.

- Intervals
  There are three "intervals" that you can define to harmonize the original sound file. If, for instance, you define interval 1 to be 400 cents, this will harmonize the sound file up a major fourth.
- Transposition Type
  If the "no. partials per interval" setting is higher than 1, you can define how the interval will be used to determine the other harmonizations. See "Spectral Filterbank" for more information on these transpositions
- Transpositions per interval
  For each interval, how many transpositions there are. For instance, 4 transpositions of a minor third, with the transposition type set to "stacked" would give you a diminished seventh chord.

SPECTRAL BIN SHIFT

Shifts the entire spectrum up or down a set number of bins. Because the bins of the FFT are equally-spaced, not logarithmically spaced, the spectrum becomes rather compressed or expanded the further away you get, and as a result, strange, inharmonic spectra result which sound quite distant from the original. Can be useful for then feeding into another effect (e.g. spectral gran, or time stretching)

- Bins to shift
  The number of bins to offset the entire spectrum
- Rotate
  Should bins that "fall off the end" be rotated around to the other side?
- Instantaneous feedback
  The bin-shifting is fed back within the same frame (as opposed to normal feedback, in which the output is fed back into the following input frame)
- Num instant feedback
  How many times the instantaneous feedback algorithm should be calculated

SPECTRAL PITCH SHIFT

A simple spectral pitch shift up or down by a set amount, using the method suggested by Mark Dolson and Jean Laroche in their paper 'New Phase-Vocoder Techniques for Real-Time Pitch-Shifting, Chorusing, Harmonizing and Other Exotic Audio Modifications', in which spectral peaks are tracked and shifted independently of one another. You can optionally use a 'Bins to Shift' parameter to shift all the spectral peaks up in an inharmonic manner.

- Pitch Shift
  The amount to shift

- Use 'Bins to Shift' parameter
  Instead of using an interval, shift by a set number of bins (this is different from the spectral bin shift, in that it shifts individual peaks by a set number of bins, rather than the entire spectrum)
- Bins to shift
  The number of bins to shift, if the above parameter is checked
- Rotate
  Should bins that 'fall off the end' be rotated around to the other end?

SPECTRAL STRETCH

An effect to create a variety of inharmonic stretches. Spectral peaks are tracked and shifted by the quadratic equation: $\beta\omega+\alpha\omega^2$, where ω is the frequency of each bin.

- Alpha stretch
  The value for α in the equation above
- Beta stretch
  The value for β in the equation above, which can be varied using the "beta stretch variance" parameter

9. SPECTRAL FILTERING EFFECTS

SPECTRAL TRACING

Typically used to retain only the n loudest bins in a spectrum and remove the others. Alternatively you can retain the n softest bins, or set a threshold level (a "gate") below which all partials are zeroed, or above which all partials are zeroed.

- Tracing type
  Choose from four different algorithms: retain the n loudest, the n softest, bins above the threshold or bins below the threshold
- Number of bins
  The number of bins to retain (either the loudest or the softest depending on the Tracing Type parameter)
- Threshold
  The threshold in dB; retains bins above or below this threshold depending on the Tracing Type parameter

SPECTRAL FILTERBANK

A bank of extremely narrow, very pure bandpass filters (similar to the GRM Reson plug-in). You can choose the interval between centre frequencies of your filterbank, and how many filters should be created. Because of the inaccuracies in pitch in the FFT, the filterbank cannot guarantee the actual frequencies you ask for will be exactly rendered; however, it will be as accurate as possible within the limitations of the FFT.

- Number of filters
  The number of filters in the filterbank
- Base frequency
  The lowest frequency in the filterbank, from which the other frequencies are built-up
- Interval

The interval that will be applied to the base frequency and successive "harmonics" to build up the filterbank, using the techniques described in the "transposition type" parameter

- Transposition type
  –Stacked: each interval is simply applied to the previous interval, "stacking" the interval on top of each other. For an interval of 300 cents (a minor third), this would create a diminished seventh chord, for instance. An interval of 200 cents would create a whole-tone scale.

  –Chord: the "fundamental" and the upper note defined by the interval are simply repeated at the octave. If the fundamental was C, and the interval was a perfect fifth (700 cents), then the notes are just C and G repeated up through the octaves

  –Harmonics: a harmonic series is based on integer multiples of a fundamental (n, 2n, 3n, 4n, etc). The harmonics setting extends this by using multiples of other intervals. It assumes that the standard harmonic series would have an interval setting of 1200 cents (an octave) – therefore, halving this to 600 cents (a tritone) would give you filters at 0.5n, n, 1.5n, 2n, etc..., while 2400 cents (two octaves) would give you 2n, 4n, 6n, 8n, etc.

- Harmonic gain
  "Harmonic gain" indicates how much each progressive "harmonic" as you get higher should be attenuated by.

- Side bins
  Allows you to create a "wider" filter by adding side bins on either side of the central filter bin. While this lets more of the original sound through, and therefore less pure filtering, it also creates some interesting "pulsating" dynamic effects

- Side bin gain
  How much each side bin is attenuated by as you get further away from the central filter

SPECTRAL GLIDING FILTERS

Creates a series of spectral filters that glide in pitch

- Number of filters
  How many filters are to be created

- Filter width
  The size of the filter in bins

- Start frequency
  The frequency at which the filter should start its life

- Start frequency variation
  Randomization of the filter Start Frequency parameter

- Pitch glide per second
  How much drift in pitch each filter should have (per second)

- Pitch glide var
  Randomization of the Pitch Glide parameter

- Filter duration
  The lifetime of each filter

- Filter duration var
  Randomization of the Filter Duration parameter

- Silence duration
  The duration of silence before a new filter is created

- Silence duration var
  Randomization of the Silence Duration parameter

10.SPECTRAL TEXTURIZING EFFECTS

SPECTRAL GRANULATION

'Grains' or 'chunks' of the spectrum, of a certain frequency range and duration are taken and delayed by a certain length. Optionally, you can also apply a bin shift and a fade in/fade out to the grains.

- Grain length
  The length of each grain. NB: the actual length is rounded down to the nearest FFT length
- Grain length variance
  Randomization of the grain length (l ± var)
- Grain delay
  The length of time to delay each group
- Grain delay variance
  Randomization of the grain delay
- Bin shift
  An optional amount to shift each grain up or down the spectrum by a number of bins
- Bin shift variance
  Randomization of the bin shift parameter
- Bins per grain
  How many bins constitute a grain. The larger this parameter the more of the original sound can be detected in the output.
- Fade amount
  Fades in and out each grain for extra smoothness. 100% is maximum fading.
- Density
  The percentage of grains that are actually delayed. 100% means the entire sound is granulation. 0% means none of the sound makes it out.

11. SPECTRAL PARTIAL EFFECTS

These effects track peaks in the spectral envelope that appear to have a certain lifetime, and creates "regions of influence" between the tracked peaks. These are regions are estimates of the stable "partials" of the sound, and so these effects work best on sounds that have a stable pitched morphology. The fuzziness parameter allows a peak to drift in pitch within a certain range of bins and still be tracked as a stable partial. If this parameter is set too small, then a pitch drift outside that boundary will end the lifetime of that partial, and it will be tracked as a new partial. On the other hand, a setting too large might cause new partials to be tracked as a drift in an existing partials. The max bins per partial allows you to reduce a 'region of influence' down to a smaller number of bins centred around the peak bin, creating 'purer' partials, a little bit like Spectral Tracing.

SPECTRAL EMERGENCE

This effect tracks the partials, but applies an amplitude envelope to them to create a strange pulsating sound. Because this happens separately for left and right channels, you often get a 'rotating' panning effect as well.

- Fade in (windows)
  The number of windows over which to fade in a tracked partial
- Hold (windows)
  Once the tracked partial has been faded in, how long should it be held at 0dB for?

- Fade out (windows)
  Once the tracked partial has been faded in and held, how long it should be faded out for?
- Recycle faded peaks
  If this is checked, then if a partial is still being tracked after it has finished all three phases of its envelope, it will be restarted from the fade in section. If unchecked, it will remain silence until the end of its life.

SPECTRAL PARTIAL GLIDE

This effect tracks partials, but shifts their pitches at a set amount. However, this amount can vary for each partial, to the point where you can get partials gliding in opposite directions!

- Pitch shift per second
  The central pitch shift, in cents, per second
- Pitch shift variance
  Randomization of the Pitch Shift per Second parameter, on a per partial basis
- Mirror bin
  A somewhat experimental parameter: bins below this number will glide in the opposite direction from bins above this number
- Fade in (windows)
  Apply a gradual fade in to tracked partials
- Zero transients
  Partials that only exist for a single window should be removed.

12. TIME DOMAIN EFFECTS

Chorus

A time-varying chorus/flanger effect

Grain Streamer

'Captures' and repeats grains

SuperFilterbank

A standard time-domain implementation of a bank of bandpass filters, with the option to choose the filter type, including comb filter, biquad, butterworth.